Innovation Project Proposal

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Introduction

Exothermic reactions release heat, which can increase the reaction rate and lead to further heat generation. This thermal feedback can result in multiple steady states and stability challenges in chemical reactors. Understanding the conditions leading to uniqueness, multiplicity, and stability is essential for safe and efficient reactor operation.

Objectives and Methodology

The project will begin with a single isothermal CSTR to establish the basic material-balance model, followed by a single non-isothermal CSTR to investigate the coupling between reaction kinetics and heat transfer. The study will then be extended to two isothermal CSTRs, two non-isothermal CSTRs, and finally a cascade of non-isothermal CSTRs. These cases will help us understand how reactor staging, heat transfer, and back-mixing influence multiple steady states and stability.

A cascade of CSTRs is not physically identical to a fixed-bed reactor; rather, it is an approximate model that represents changes in concentration and temperature along the bed through a series of ideally mixed stages.

Proposed Extension(if time permits)

Building on the baseline analysis, the project will investigate whether distributing a fixed total cooling capacity non-uniformly across a CSTR cascade can improve reactor performance compared with uniform cooling. The study will examine the effects of cooling distribution on peak reactor temperature, overall conversion, the number of steady states, and the range of operating conditions exhibiting multiplicity.

Expected Outcomes

The project aims to develop an understanding of multiplicity and stability in non-isothermal reactor systems and evaluate whether optimising cooling distribution can reduce peak temperatures and the range of operating conditions exhibiting multiplicity while maintaining acceptable conversion.

Reference: Bremer, J. and Sundmacher, K. (2021). *Novel Multiplicity and Stability Criteria for Non-Isothermal Fixed-Bed Reactors.* Frontiers in Energy Research, 8, 549298. https://doi.org/10.3389/fenrg.2020.549298